系統程式設計
Lab3 Demo Tables

學號:413410110
姓名:吳俊霆

Demo Part3

BUFFSIZE	file name	1mb_file	50mb_file	500mb_file
100	real	0m0.026s	0m0.939s	0m10.495s
	user	0m0.009s	0m0.185s	0m1.902s
	sys	0m0.022s	0m0.749s	0m7.868s
4096	real	0m0.006s	0m0.036s	0m0.383s
	user	0m0.001s	0m0.006s	0m0.047s
	sys	0m0.002s	0m0.027s	0m0.307s
8192	real	0m0.004s	0m0.026s	0m0.285s
	user	0m0.001s	0m0.004s	0m0.024s
	sys	0m0.002s	0m0.017s	0m0.219s
16384	real	0m0.005s	0m0.025s	0m0.243s
	user	0m0.001s	0m0.003s	0m0.013s
	sys	0m0.002s	0m0.014s	0m0.135s

```
lk@LK Lab3_example % ./test.sh
BUFFSIZE: 100
1mb

real	0m0.026s
user	0m0.009s
sys	0m0.022s


BUFFSIZE: 4096
1mb

real	0m0.006s
user	0m0.001s
sys	0m0.002s


BUFFSIZE: 8192
1mb

real	0m0.004s
user	0m0.001s
sys	0m0.002s


BUFFSIZE: 16384
1mb

real	0m0.005s
user	0m0.001s
sys	0m0.002s


BUFFSIZE: 100
50mb

real	0m0.939s
user	0m0.185s
sys	0m0.749s


BUFFSIZE: 4096
50mb

real	0m0.036s
user	0m0.006s
sys	0m0.027s
BUFFSIZE: 8192
50mb

real	0m0.026s
user	0m0.004s
sys	0m0.017s


BUFFSIZE: 16384
50mb

real	0m0.025s
user	0m0.003s
sys	0m0.014s


BUFFSIZE: 100
500mb

real	0m10.495s
user	0m1.902s
sys	0m7.868s


BUFFSIZE: 4096
500mb

real	0m0.383s
user	0m0.047s
sys	0m0.307s


BUFFSIZE: 8192
500mb

real	0m0.285s
user	0m0.024s
sys	0m0.219s


BUFFSIZE: 16384
500mb

real	0m0.243s
user	0m0.013s
sys	0m0.135s
```

Demo Part4

BUFFSIZE	file name	512mb_file1	512mb_file2	512mb_file2
4096	real	0m0.426s	0m0.444s	0m0.292s
	user	0m0.051s	0m0.049s	0m0.047s
	sys	0m0.331s	0m0.349s	0m0.241s

```
Part 4

real    0m0.426s
user    0m0.051s
sys     0m0.331s

real    0m0.444s
user    0m0.049s
sys     0m0.349s

real    0m0.292s
user    0m0.047s
sys     0m0.241s
```

Demo Part5

BUFFSIZE	file name	512mb_file3	512mb_file4	512mb_file4
4096	real	0m1.064s	0m0.498s	0m0.358s
	user	0m0.075s	0m0.071s	0m0.070s
	sys	0m0.377s	0m0.378s	0m0.284s

```
Part 5

real	0m1.064s
user	0m0.075s
sys	0m0.377s

real	0m0.498s
user	0m0.071s
sys	0m0.378s

real	0m0.358s
user	0m0.070s
sys	0m0.284s
```

1. How did you direct the output file to /dev/null?

   By using the open system call with “0_WRONLY” flag on path “/dev/null”.

2. In the table of Part3, do bigger buffer sizes make IO faster?

   Yes, by increasing buffer size the IO times decrease, but only to a certain degree

3. In the table of Part4, what did you observe? Why?

   The second time running “time ./buffsize4096 < 512mb_file2”, the real time was lower than the first run.

   Due to linux page cache, in the first run the data is being copied from the disk to RAM, but on the second run the OS realizes that the data’s already in RAM hence improving real time speed.

4. In the table of Part5, what did you observe? Why?

   The real time increased a lot and the caching from Part4 didn’t happen.

   When fsync is in the loop, it forces the OS to pause and wait till the data is stored on the physical disk instead of

telling the program that the write is done when the data's stored in RAM.